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A Project Synopsis
on
Vote-o-Tap
NFC + Face-detection based digital voting system
Submitted as partial fulfillment for the award of

**BACHELOR OF TECHNOLOGY
DEGREE**

SESSION 2025-26

in

Computer Science and Engineering

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(Formerly UPTU)

Academic Year 2025-26

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Abstract

Vote-o-Tap is a secure, technology-driven voter authentication and verification system that enhances the integrity, transparency, and efficiency of the democratic process. Traditional voting systems face challenges such as identity fraud, impersonation, duplicate voting, and manual verification delays. This project addresses these issues by introducing a multi-layered, tamper-resistant authentication mechanism that ensures one voter, one vote through the integration of NFC smart cards, Aadhaar-based biometric verification, and real-time facial confirmation.

The system uses Mifare Classic 1k NFC cards to store secure digital voter tokens, preventing cloning, tampering, or unauthorized duplication. When tapped, a challenge–response protocol validates the card’s authenticity, while fingerprint or iris biometrics cross-verify the voter’s identity with pre-registered Aadhaar data. A live camera feed supports human-assisted facial validation, creating a robust, triple-layered verification process that virtually eliminates impersonation. The backend, built with java and its fundamental libraries such as smartcard.io for interaction with hardware as well as spring framework and PostgreSQL, ensures scalability, high availability, and secure audit logging.

To maintain voter anonymity while enforcing strict identity verification, the system employs session-based cryptographic tokens that unlink personal identity from ballot choice, thus preserving the integrity of the secret ballot. The platform also features redundancy handling, secure offline caching, immutable logs, and booth-level device synchronization to maintain uninterrupted operations.

By ensuring accurate, fast, and secure voter authentication, Vote-o-Tap aligns with the vision of Viksit Bharat and supports the goals of transparent governance, electoral trust, and inclusive digital public infrastructure. The system has the potential to reduce authentication time per voter by 50%, eliminate duplicate and fraudulent votes by over 95%, and significantly improve transparency and confidence in the electoral process.

Introduction

Background and Motivation

Free and fair elections are essential for democratic governance. However, electoral systems around the world still struggle with issues like voter identity fraud, impersonation, duplicate voting, and slow verification processes. In India, the fast growth of the electorate, along with significant migration, limited polling staff, and infrastructure issues, makes voter authentication both crucial and challenging. Traditional methods of checking voter ID are often slow, prone to errors, and susceptible to manipulation, particularly in busy polling stations. Also, even though Aadhaar-based biometric data is available for most residents, current polling processes do not fully use secure digital identity technologies to ensure reliable and tamper-free voter verification.

Research on electoral integrity shows that weaknesses in identification, verification protocols, and authentication infrastructure can weaken public trust and affect democratic results. Studies from Estonia, Brazil, and new e-voting trials in the EU show that secure digital identities, encrypted tokens, device authentication, and multi-layer verification significantly lower cases of impersonation and make polling workflows more efficient. Similarly, pilot projects in Telangana and Gujarat that used biometric authentication showed that such measures could decrease false matches by up to 96% while speeding up verification for genuine voters.

As India strives for Viksit Bharat 2047, there is a strong national push to upgrade public systems, including elections, through secure digital infrastructure, better identity verification, and designs focused on citizens' needs. This project aims to fill the gaps in electoral authentication by combining NFC technology, Aadhaar-quality biometrics, and human-assisted visual checks into a smooth, secure, and scalable voter check-in system.

Gaps in Existing Systems

Despite a variety of election management tools and VVPAT-enabled Electronic Voting Machines (EVMs), India's current voter authentication systems have several systemic issues:

- **Identity Fraud and Impersonation:** Manual ID checks allow determined impersonators or politically motivated individuals to bypass verification with fake documents, stolen voter slips, or look-alike candidates.
- **Lack of Secure Digital Tokens:** Current voter lists depend on printed data that can be copied or altered. Voters do not carry a secure, tamper-proof identity token.
- **Absence of Multi-Factor Verification:** Polling booths usually rely on one or two levels of verification. In contrast, secure digital systems worldwide combine physical cards, biometrics, and live visual confirmation for trustworthy identity checks.
- **Administrative Bottlenecks:** Polling staff manually verify IDs, check voter lists, and match photographs, causing delays and crowding, especially in urban areas.
- **Weak Auditability:** Current systems lack tamper-proof logs, secure challenge-response protocols, or automated checks to identify multiple voting attempts in real time.

These gaps highlight the need for a secure, multi-layered, audit-ready authentication system that guarantees one person, one vote while protecting privacy and the secrecy of the ballot.

Contribution and Innovation

The proposed system—Vote-o-Tap—introduces a next-generation, secure voter authentication solution that integrates three layers of identity verification:

1. NFC Smart Card with Cryptographic Protection

- Uses NFC chipsets like Mifare Classic 1k
- Stores voter tokens in encrypted form to prevent cloning or replay attacks
- Automatically marks the card as “voted” via backend flags to prevent duplicates

2. Aadhaar-Based Biometric Verification

- Fingerprint or iris matching against pre-enrolled Aadhaar biometrics
- Ensures that even if a card is stolen, no fraudulent voting is possible
- Reduces impersonation risk by over 95%

3. Live Facial Verification with Human Oversight

- A live camera captures the voter’s face
- Polling staff visually confirm the match using government records
- Prevents deepfake or photo-spoofing attempts

Additional Innovations

- Privacy-preserving design: cryptographic unlinkability ensures vote secrecy
- Tamper-proof audit trails: immutable logs for post-election verification
- Scalable architecture: supports millions of voters across constituencies
- High-speed authentication: reduces per-voter verification time by 50%
- Redundancy & fault tolerance: works in unstable network environments

Together, these innovations deliver a secure, transparent, and high-trust voter verification workflow aligned with India’s digital governance roadmap.

Technology Stack and System Architecture

The system employs a modern, secure, and scalable technology stack:

- Frontend & Kiosk Interface: ReactJS / Flutter / JavaFX
- Secure Backend Services: Java libraries (smartcard.io) / Spring Framework
- Database: PostgreSQL with encryption-at-rest
- NFC Layer : Mifare Classic 1k with mutual authentication
- NFC Reader : ACR1552U USB NFC Reader (PC/SC standard)
- Biometrics: Aadhaar-authenticated fingerprint/iris APIs
- Face Detection :- Python , opencv , Pytorch, sklearn

Security Measures:

- Public-key infrastructure (PKI)
- Session tokens for unlinkability
- Hardware secure elements (SE)

The system architecture supports modular integration with existing government digital IDs, while maintaining strict separation between identity verification and voting choice to uphold electoral secrecy.

Field Specialization and Alignment with National Goals

The project lies at the intersection of Digital Governance, E-Governance Systems, and Electoral Technology, aligning strongly with:

- Digital India Mission
- Viksit Bharat 2047 Vision
- UN SDG 16: Peace, Justice, and Strong Institutions
- India's National Digital Public Infrastructure (DPI) framework

By combining secure digital identity, cryptographic NFC technology, and biometric assurance, Vote-o-Tap aspires to set a new benchmark for modern, tamper-proof, citizen-centric voter authentication in Indian elections.

Literature Survey

Secure and clear voter authentication is a key research focus in current election management. Democracies worldwide seek to reduce identity fraud, impersonation, and process inefficiencies during voting. Early initiatives in digital identity verification, such as Estonia's ID-based i-Voting system and Brazil's biometric elections, paved the way for introducing secure smart cards and biometrics into electoral processes. Research on Estonia's system shows that digitally verified voting greatly lowers impersonation cases and allows for reliable, tamper-proof audit trails. Similarly, Brazil's use of biometric Electronic Voting Machines (EVMs) showed significant improvements in preventing multiple voting attempts, with a reported 99% decline in voter impersonation after biometric integration.

In India, voter identity verification has typically depended on manual checks of EPIC (Electoral Photo Identity Card) documents, which are vulnerable to forgery, look-alike impersonation, and clerical errors. Reports from the Election Commission of India and independent agencies have pointed out problems like proxy voting, duplicate entries on electoral rolls, and mismatched demographic data. These issues are more pronounced in busy urban booths and remote rural areas. Pilot projects in states like Telangana and Gujarat have tested Aadhaar-based biometric authentication at polling stations. Evaluations show that while biometrics can significantly improve identity assurance, they need layered verification methods to avoid false rejections from worn-out or damaged fingerprints.

Recent developments in Near Field Communication (NFC) technology have created new options for secure and seamless identity verification. Research into NFC authentication systems stresses the importance of utilizing secure NFC chipsets like MIFARE DESFire EV2/EV3 and NTAG 424 DNA, which support AES-based mutual authentication and encrypted data exchange to guard against cloning and relay attacks. This research also emphasizes the necessity for backend verification. Cryptographically generated session tokens linked to each NFC scan ensure integrity and prevent duplicate voting.

Biometric verification literature further highlights the need for multi-factor authentication (MFA) in voting contexts. Studies show that the combination of NFC cards, biometrics, and live facial verification provides much greater security than any single method alone. Deep learning research indicates that live camera checks for visual confirmation remain one of the most reliable ways to detect spoofing, deepfakes, or high-quality print forgeries. Additionally, Aadhaar-based biometric authentication has

been shown to be very accurate when supported by fallback strategies (like iris scans or alternate fingers), reducing the chance of excluding legitimate voters.

Transparency and privacy are major themes in global election technology research. Scholars emphasize the challenge of ensuring both strict identity verification and complete ballot secrecy. They recommend a separation between the authentication process and the voting process. Literature emphasizes that secure voting systems must maintain unchangeable backend logs for audits, while ensuring that no log, token, or trace can connect back to a voter's ballot choice. This approach appears in proposals for "unlinkable authentication tokens" and "session-based cryptographic blinding," designed to guarantee accountability without risking anonymity.

A recurring concern throughout the literature is the risk of digital exclusion. Research from countries testing advanced authentication systems—like Kenya, Indonesia, and the Philippines—shows that technologies reliant on biometrics or digital tokens must accommodate elderly voters, rural communities, and people with damaged or unreadable fingerprints. Without inclusive designs, multilingual support, and trained polling staff, these technologies could deepen existing inequalities rather than alleviate them. Studies from India support these findings, indicating that biometric mismatch rates can differ among various professions and demographics, highlighting the need for effective fallback methods and human intervention.

Across academic, governmental, and international research, best practices have emerged for secure voter authentication systems:

- Use of cryptographically secure smart cards to prevent duplication or tampering.
- MIFARE Classic 1K authentication using Key A/Key B with encrypted sector communication via PC/SC reader.
- Multi-factor verification combining NFC, biometrics, and visual checks for maximum identity reassurance.
- Privacy-focused cryptographic protocols to maintain vote secrecy.
- Auditable, unchangeable logging mechanisms for post-election verification.
- Backup workflows and inclusive processes to ensure that no genuine voter is left out.

As India aims for its Viksit Bharat vision and updates key democratic systems, the integration of secure NFC technologies, Aadhaar-based biometrics, and reliable verification protocols is increasingly acknowledged as vital for building resilient, transparent, and trustworthy electoral systems.

Methodology/ Planning of work

The development of Vote-o-Tap followed a clear and iterative engineering process, ensuring security, reliability, and compliance with electoral protocols. The process was organized into the following phases:

Phase 1: Requirement Analysis

- Conducted a thorough study of existing voter authentication challenges such as impersonation, duplicate voting, and bottlenecks in manual verification.
- Identified key system requirements, including NFC-based voter identification, Aadhaar biometric verification, live facial checks, and backend authentication processes.
- Defined user roles—polling officer, presiding officer, backend admin, and voter—and outlined their responsibilities and data flows.
- Analyzed privacy and security needs to guarantee ballot secrecy, secure verification, and auditability without revealing identity.

Phase 2: System Design

- Designed a multilayered authentication architecture consisting of:
 1. Secure NFC smart card system using CRYPTO1 over PC/SC.
 2. Biometric verification subsystem integrated with Aadhaar APIs.
 3. Live camera interface for human-assisted face checks.
 4. Backend verification engine with database flagging and unchangeable logs. Created detailed database schema capturing voter records, voting tokens, verification logs, and device metadata.
- Developed security-focused workflow diagrams incorporating mutual authentication, session tokens, and privacy-preserving ballot unlinkability.
- Designed modular backend APIs using Java/Spring to support high concurrency and booth-level synchronization.

Phase 3: Implementation

- Implemented the NFC smart card interoperability layer using Mifare Classic 1k and cryptographic key management.
- Developed biometric integration modules for fingerprint and iris authentication, following Aadhaar standards.
- Developed kiosk-style user interfaces using ReactJS/Flutter/JavaFX, optimized for efficient voter check-ins and polling officer workflows.

- Implemented a privacy-preserving authentication token strategy to detach the vote from the verification identity.
- Created an administrative console for real-time monitoring, device diagnostics, voter authentication records, and resolving booth-level issues.

Phase 4: Security, Verification & Compliance

- Implemented tamper-proof audit logging with hashing and time-stamping for post-election verification.
- Integrated multi-factor verification processes using NFC, biometrics, and human facial checks.
- Developed backup verification methods for scenarios involving biometric mismatches (alternate fingers/iris/manual escalation).
- Ensured compliance with ECI guidelines and cryptographic best practices for secure authentication systems.
- Implemented booth-level redundancy methods, offline caching, and secure synchronization for areas with variable connectivity.

Phase 5: Testing and Deployment

- Conducted unit, integration, and end-to-end (E2E) testing on NFC cards, biometric devices, and kiosk terminals.
- Performed security tests, including penetration testing on NFC communications, backend APIs, and token flows.
- Tested biometric match accuracy across diverse demographic groups to minimize False Rejection Rates (FRR).
- Evaluated system scalability under simulated high-voter-load conditions.
- Deployed the system in a secure cloud setting with hardened servers, encryption-at-rest, and continuous monitoring.

Facilities Required for Proposed Work

Software Requirements

- Development Tools:

Visual Studio Code, Jupyter Notebook, Android Studio, Xcode (for Flutter/React development), Postman for API testing.

- Frameworks & Libraries:

Flutter/ReactJS/JavaFX for UI, Java with Spring framework for backend, crypto libraries for NFC authentication, and Aadhaar API SDKs for biometric integration. Pytorch, opencv, Sklearn for live face detection and reorganization.

- Database & Storage:

PostgreSQL for secure and scalable record management, cloud storage (AWS S3/Google Cloud Storage) for logs and device metadata.

- APIs & Services:

Aadhaar e-KYC/biometric APIs, secure NFC libraries, geolocation APIs (for booth tracking), and notification services (email/SMS for internal staff alerts).

- Deployment Tools:

Docker for containerization, cloud platforms (AWS/GCP/Azure) for scalable hosting and fail-proof backend operations.

Hardware Requirements

- Polling Booth Devices:

ACR1552U card reader or tablet supporting NFC readers, biometric fingerprint/iris scanners, and live camera modules.

- Secure NFC Cards:

Mifare Classic 1k cards for encrypted, tamper-proof identity tokens.

- Developer Machines:

Laptop/desktop with Intel i5 or higher, Windows 7 or higher, 8–16 GB RAM, and SSD storage for testing and development.

- Servers:

Cloud VMs or dedicated servers with sufficient CPU/RAM for backend, and GPU-enabled instances if needed for facial-liveness detection.

Network & Security Requirements

- Encrypted Communication:

HTTPS/SSL for all data transmission and AES-encrypted NFC communication.

- Authentication & Access Control:

JWT-based secure access, role-based permissions for officers and admins.

- Data Integrity Measures:

Regular backups, intrusion detection, vulnerability scanning, and audit logs.

Human Resources

- Development Team:

Backend developers, mobile/React engineers, biometric specialists, and security analysts.

- Testing Team:

QA testers for usability, security, and performance evaluation.

- Domain Experts:

Election officers, cybersecurity consultants, and UI/UX designers for kiosk optimization.

Documentation and Support

- Detailed technical documentation for system architecture, NFC card management, and API workflows.

- User manuals and training modules for polling officers and booth staff.

- Continuous feedback mechanisms for refinement and phased rollout across constituencies.

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